STACKS, DISCRIMINANTS: LEMMA-COMPARE-TRACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3433-3447 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 3449-3492, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_discriminant.json (pb=1777, stmt 3433-3447, proof 3449-3492).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AWZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 3492–3492. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A9Y: sources/stacks-discriminant/pinned-original-source.tex, source lines 3492–3492. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BUL: sources/stacks-discriminant/pinned-original-source.tex, source lines 3360–3375. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT8: sources/stacks-discriminant/pinned-original-source.tex, source lines 874–877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVJ: sources/stacks-discriminant/pinned-original-source.tex, source lines 1107–1124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVH: sources/stacks-discriminant/pinned-original-source.tex, source lines 694–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : Y \to X$ be a finite flat morphism of Noetherian schemes. The map*

$$\mathit{Trace}_f : f_*\mathcal{O}_Y \longrightarrow \mathcal{O}_X$$

of Section 0BVH corresponds to a map $\mathcal{O}_Y \to f^!\mathcal{O}_X$ (see proof). Denote $\tau_{Y/X} \in H^0(Y, f^!\mathcal{O}_X)$ the image of 1. Via the isomorphism $H^0(f^!\mathcal{O}_X) = \omega_{X/Y}$ of Lemma 0BUL this agrees with the construction in Remark 0BVJ.

Proof. The functor $f^!$ is defined in Duality for Schemes, Section 0A9Y. Since f is finite (and hence proper), we see that $f^!$ is given by the right adjoint to pushforward for f. In Duality for Schemes, Section 0AWZ we have made this adjoint explicit. In particular, the object $f^!\mathcal{O}_X$ consists of a single cohomology sheaf placed in degree 0 and for this sheaf we have

$$f_*f^!\mathcal{O}_X = \mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X)$$

To see this we use also that $f_*\mathcal{O}_Y$ is finite locally free as f is a finite flat morphism of Noetherian schemes and hence all higher Ext sheaves are zero. Some details omitted. Thus finally

$$\mathrm{Trace}_f \in \mathrm{Hom}_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X) = \Gamma(X, f_*f^!\mathcal{O}_X) = \Gamma(Y, f^!\mathcal{O}_X)$$

On the other hand, we have $f^!\mathcal{O}_X = \omega_{Y/X}$ by the identification of Lemma 0BUL. Thus we now have two elements, namely Trace_f and $\tau_{Y/X}$ from Remark 0BVJ in

$$\Gamma(Y, f^!\mathcal{O}_X) = \Gamma(Y, \omega_{Y/X})$$

and the lemma says these elements are the same.

Let $U = \mathrm{Spec}(A) \subset X$ be an affine open with inverse image $V = \mathrm{Spec}(B) \subset Y$. Since f is finite, we see that $A \to B$ is finite and hence the $\omega_{Y/X}(V) = \mathrm{Hom}_A(B, A)$ by construction and this isomorphism agrees with the identification of $f_*f^!\mathcal{O}_Y$ with $\mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X)$ discussed above. Hence the agreement of Trace_f and $\tau_{Y/X}$ follows from the fact that $\tau_{B/A} = \mathrm{Trace}_{B/A}$ by Lemma 0BT8. □